

Design and function

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General structure of the machine

Base frame	The machine is mounted on a base frame made of stainless steel with vibration-damping and variable-height supports.
Panelling	The machine is equipped with protective doors with protective switches and with stainless steel doors with locking levers in the areas for drive and installation. For further information on the safety devices, please refer to the Chapter "Safety".
Production, drive and installation areas	The production area is separated from the drive and installation areas by means of a plate. This means that, depending on the machine type, the machine's electric and pneumatic equipment is housed in the rear or lower part of the machine.

Limits of the machine



Attention!

If hazardous, flammable/inflammable materials are processed, the system owner is responsible for:

- safe processing,
- extraction device in the working area.

Fundamentals of the machine design

Principle The machine/system is constructed according to the state of technology and the recognised safety rules. Nevertheless, when used, dangers may arise for the life and limb of the user or third parties or the machine and other property may be impaired.

Using the machine The machine must only be operated in perfect technical working order and in the way intended, with attention given to safety and potential dangers whilst observing the operating instructions! In particular, immediately rectify (or have rectified) malfunctions which could affect safety!

Intended use

Application of the machine The machine is suitable for assembling tips and caps together and for sealing them along with infed catheters in film.

The machine may only be operated as intended with the products listed in the corresponding business order and/or formats. Only personnel who are appropriately qualified to perform their respective tasks may work at the machine (see operating personnel / user groups).

Any other use is considered as non-intended use. The manufacturer accepts no liability for damage resulting as a consequence of this; the system owner bears the associated risk.

The intended use includes the adherence to the operating, maintenance and repair conditions specified by the manufacturer.

Residual risk

In spite of all precautions, the following residual risk remains:



Attention!

Risk of injury.

Protective doors are bridged when in Setup operation. As a result, it is possible to perform machine movements while the safety doors are open. Only authorised and trained persons who are familiar with the machine operation and are aware of the possible dangers may bridge protective doors.

- In Setup operation, the machine must be operated by only one person.
- Secure the machine against accidental operation by another person, e.g. affix warning sign.



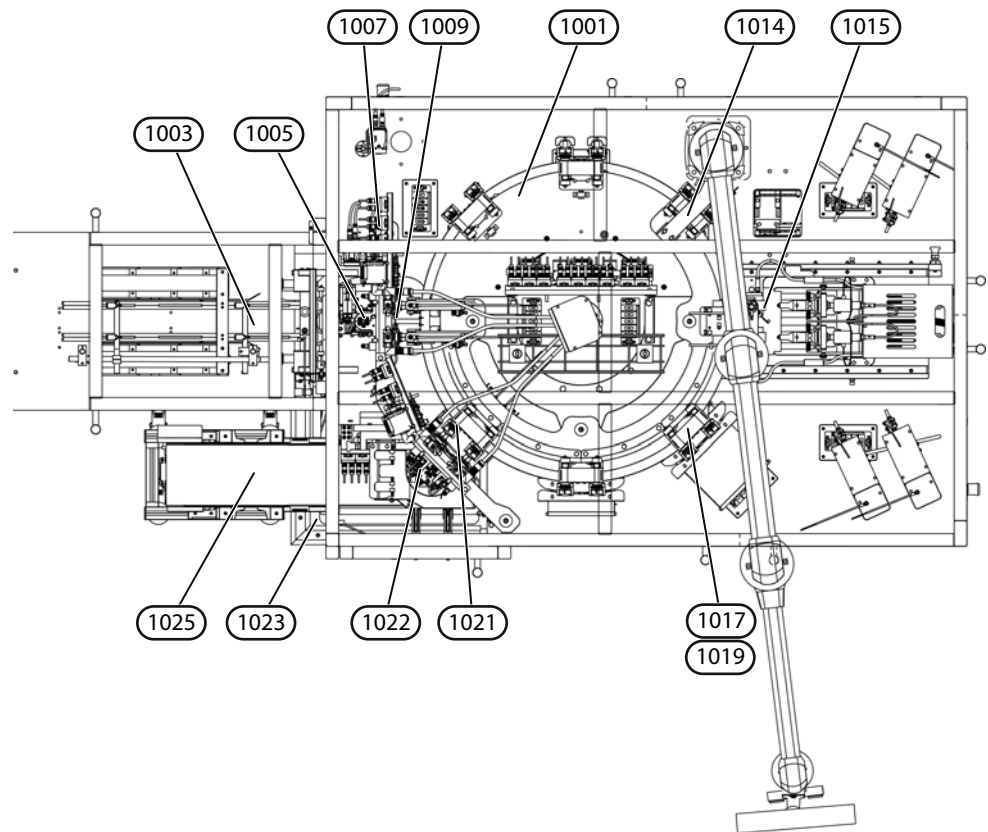
Warning!

Risk of crushing.

Tools can crush hands and fingers.

Do not reach in between the station tools.

Machine overview



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The machine consists of the following stations:

- Station 1001 - Rotary Turret
- Station 1003 - Catheter Feeding
- Station 1005 - Catheter Transfer in
- Station 1007 - Catheter Verification
- Station 1009 - Catheter load onto mandrels
- Station 1015 - Eye Punching
- Station 1017 - Vision system front
- Station 1019 - Vision system rear
- Station 1021 - Catheter unload from Mandrel



Station 1022 - Catheter Transfer out

Station 1023 - Reject

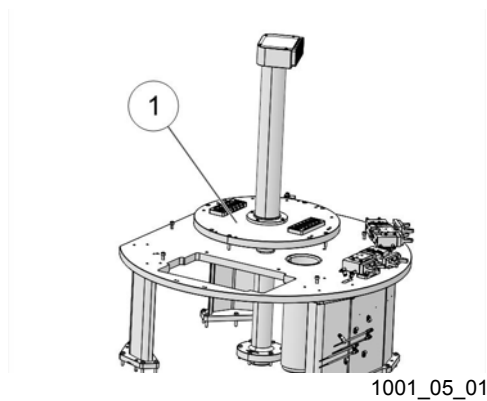
Station 1025 - Outfeed

Functional description in regular operation

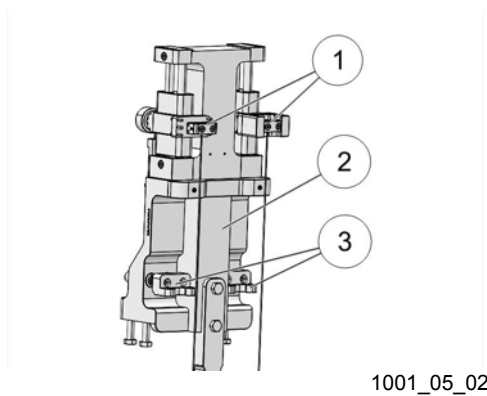
Station 1001 - Rotary turret

Mounted on the frame is a turntable (1). The turntable is the central drive unit of the system. A servo motor moves the turntable.

Mounted on the turntable (1) are eight support turrets (2). Each support turret offers space for two mandrels. Mandrels and catheters are transported by the support turrets through the system. Grippers (3) hold the catheters on the funnel.



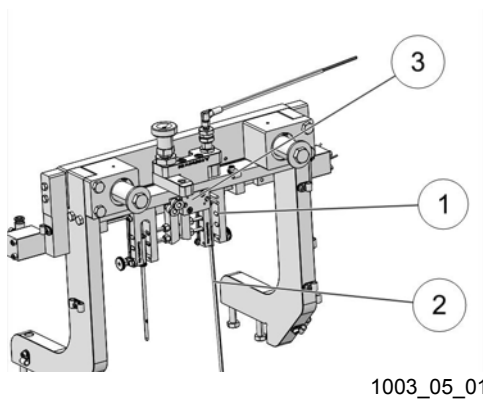
- Turntable (1)



- Mandrel (1)
- Support turret (2)
- Gripper (3)

Station 1003 - Catheter feeding

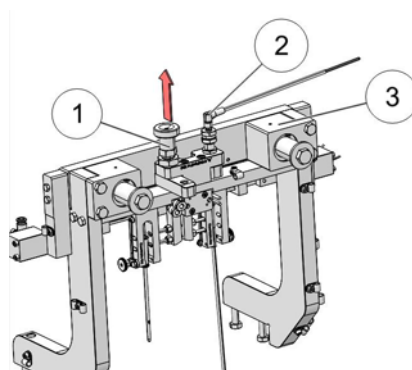
The catheters are continually conveyed through the system in two lanes. The system, however, is only capable of processing the catheters in cycles. A lock slide blocks the catheter infeed until a gripper is ready to collect the catheter. A gripper with centring pin separates the catheters. The lock slide moves aside. A catheter glides into the gripper. The gripper closes. The lock slide moves to its home position and thereby blocks the path for the following catheters. The centring pins retract and allow one catheter through before blocking the infeed again.



- Lock slide (1)
- Catheter (2)
- Gripper with centring pin (3)

Operation and settings

If the catheters are not fed in precisely, congestion can build up. To clear any congestion easier, the separator can be pushed apart.



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- Locking bolt (1)
- Sensor (2)
- Bar (3)

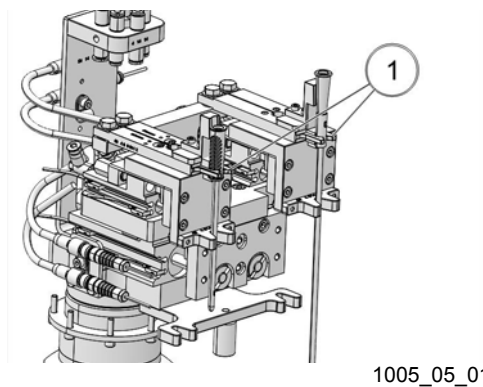
Pushing separator apart:

- ◆ Pull up the locking bolt (1) .
- The separator is released.
- ◆ Push the rail (2) with the lock pusher to the centre position of the system.
- ◆ Remove the congestion.
- ◆ Push the rail (2) with the lock pusher to the home position.
- ◆ Allow the locking bolt (1) to engage.
- ◆ The sensor (3) monitors the position of the rail (2) with lock pusher.

Station 1005 - Catheter transfer in

Two grippers collect one catheter each from the separator. A servo motor swivels the grippers with catheters 90° into the testing station.

After checking, the grippers are swivelled a further 90° towards the catheter drawing unit.

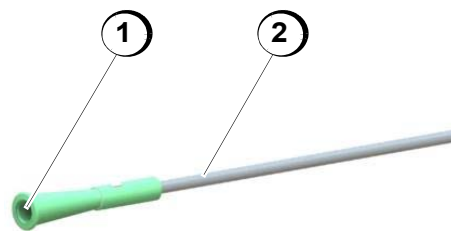


- Gripper (1)

Station 1007 - Catheter verification

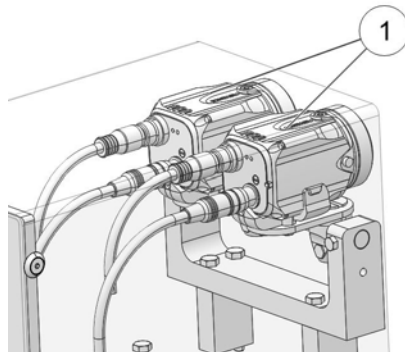
The catheter infeed swivel unit positions the catheters in front of the cameras. The cameras then detect the colour of the funnels.

If the colour is not correct, the system stops. The faulty catheter must be manually removed.



Catheter

- Funnel (1)
- Tubing (2)



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- Cameras (1) for colour detection



Observe!

For information about operation, function and adjustment, please refer to the manufacturer's operating instructions.

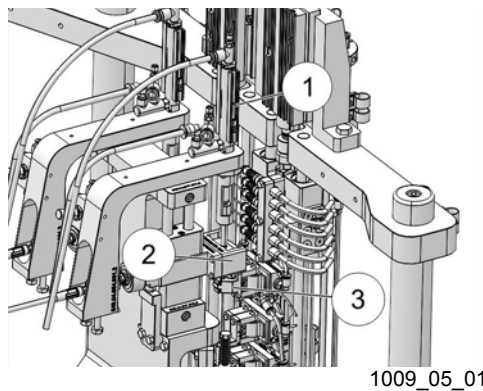
Station 1009 - Catheter load onto mandrels

The catheter infeed swivels the catheters under the mandrels.
The gripper (1) takes the catheter.

The grippers (2) of the catheter inserting unit collect the catheter in the bottom position and push the catheters onto the mandrels as they move up.

To prevent the mandrels from being damaged during fitting, a force limiter (3) is installed above the mandrels. If the wire of the mandrel strikes a kink in the catheter during insertion, the wire is raised. The force limiter reports this and the system stops.

The faulty catheter must be removed.



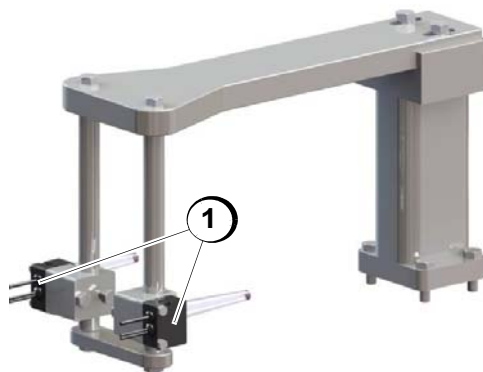
- Force limiter (1)
- Gripper (2)
- Gripper (3)

Station 1014 - Check colour of mandrels

There are catheters of different lengths and diameter. If the diameter changes, suitable mandrels must be inserted. Each size of mandrel has a different colour.

In station 1014, the colour of the mandrels is checked and so, too, is the size.

If the colour is correct, the mandrels are conveyed to the next station. If the colour is incorrect, the support turret must be clocked further until it reaches station 1013. The mandrels are checked there for straightness.



- Sensors (1) check the colour of the mandrels.

Mandrel_colour_2

Station 1015 - Eye punching



Warning!

Warning - magnetic field.

Electronic devices (pacemakers) may be affected or damaged.

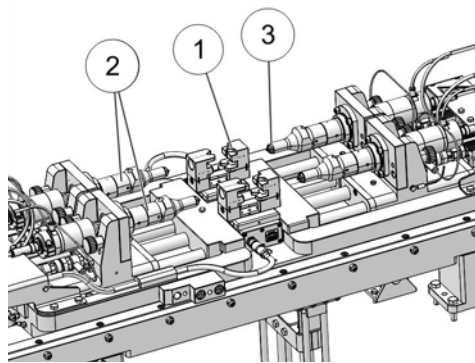
- Persons with pacemakers must keep away from the machine.
- Persons with pacemakers must not work on the machine.
- Sensitive electronic devices must not be brought near the LinMot (linear motor).
- Magnetic data carriers must not be brought near the LinMot (linear motor).

Catheters and mandrels swivel with the support turret into the processing position. A parallel gripper (1) secures the catheter together with the mandrel.

Four ultrasonic units (2) move to the working position and punch two holes into each catheter. Punching tools (3) are mounted on the ultrasonic units. These determine the size and shape of the holes.

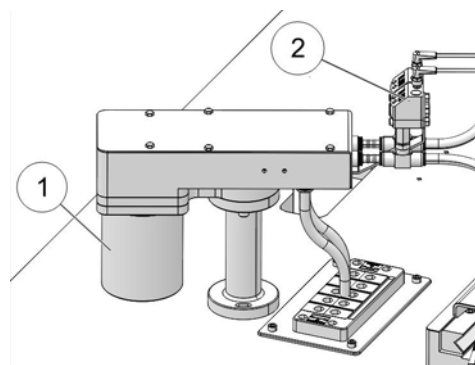
An extractor is integrated in the ultrasonic unit. This extractor collects the waste from punching and sends it via hoses to a waste bin (4).

Sensors are located along the path to the waste bin (4) and these monitor whether the waste passes by. If this is not the case, the control system is alerted and the catheter is automatically removed at the end of the system.



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- Parallel gripper (1)
- Ultrasonic units (2)
- Punching die (3)



1015_05_02

- Waste container (1)
- Sensor for monitoring whether punching waste is extracted (2)

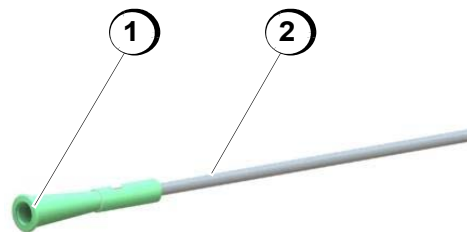
Station 1017/1019 - Vision system front/rear

The catheter coating is checked in the station and whether punching in the catheter was carried out correctly. In order to check punching in the catheter, the mandrel in front of the station is pulled out from the catheter slightly.

There must be two punches in each catheter of a different height.

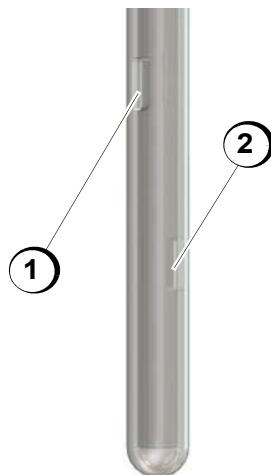
The lower punch is checked by the camera in station 1017. The upper punch is checked by the camera in station 1019.

In station 1021 and 1023, the good catheters are forwarded for further processing and the bad catheters are removed from the system.



- Funnel (1)
- Tubing (2)

Catheter

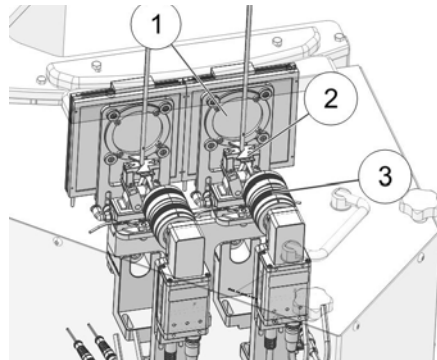


- Upper punch (1)
- Lower punch (2)

Catheter punches

Station 1017

Checks the lower punch

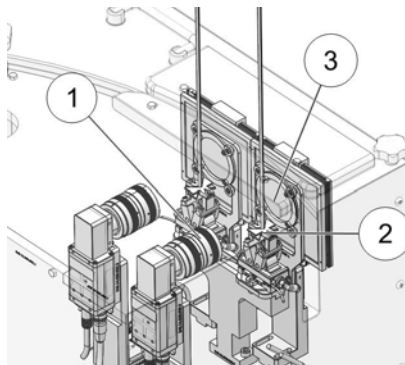


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- Backlight (1)
- Gripper (2)
- Camera (3)

Station 1019

Checks the upper punch



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- Camera (1)
- Gripper (2)
- Backlight (3)



Observe!

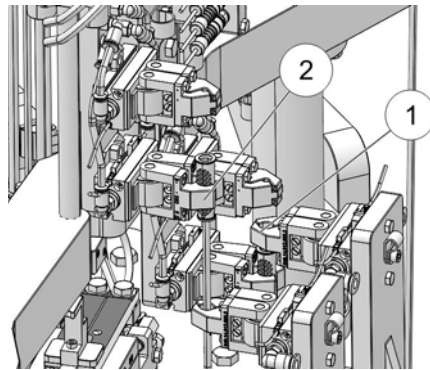
For information about operation, function and adjustment, please refer to the manufacturer's operating instructions.

Station 1021 - Catheter unload from mandrel

At this station, the catheters are pulled off from the mandrels. The value for lifting the grippers (1) comes from a sensor of the catheter infeed station.

The gripper (2) takes the catheters. The grippers (1) move back up.

The gripper of the sleever takes the catheters and swivels them out of the system. The catheters are forwarded depending on the operating sequence in the program. They are either passed on to the sleever, the conveyor belt or the waste bin.

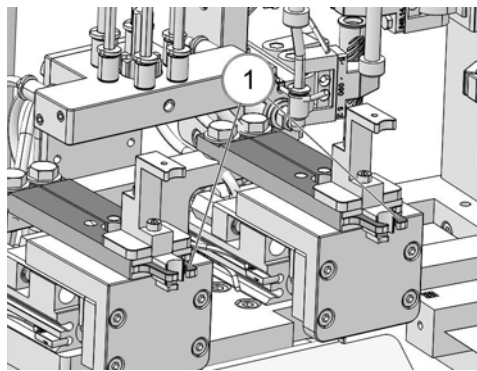


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- Gripper (1)
- Gripper (2)

Station 1022 - Catheter transfer out

The grippers (1) take the catheters out of the system. A servo motor swivels the grippers to the required position. This could be the position for transfer to the sleeve or for transfer to the conveyor belt. The grippers pass on the catheters and swivel back to the withdrawal position.

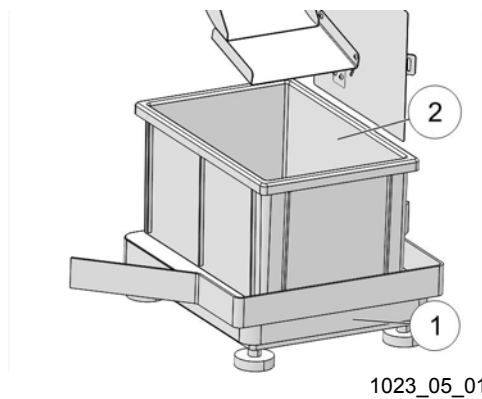


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- Gripper (1)

Station 1023 - Reject

Scales and stacking box are located beneath the catheter pull off unit. If faulty catheter arrive in the station, the grippers drops these into the stacking box directly underneath. The scales determine the weight and send the data to the control system. Once the stacking box is full, the scales submit a report. The stacking box must then be exchanged or emptied.



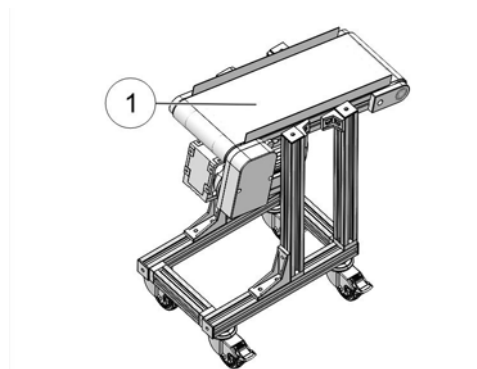
- Scales (1)
- Stacking box (2)

Station 1025 - Outfeed

Once processing is complete, the catheters can either be conveyed to the sleeve or onto a conveyor belt.

Grippers remove the catheters from the system, swivel to the conveyor belt and allow them to drop onto the conveyor belt.

The conveyor belt is attached to the catheter application unit. For better operation, it can be pushed aside quickly and easily.



1025_05_01

- Conveyor belt (1)